

Harish Vijayakumar

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Education

Georgia Institute of Technology | Atlanta, GA

Bachelor of Science in Computer Engineering

Expected Graduation, May 2028

GPA 3.73/4.0

Relevant Coursework: Digital System Design, Circuit Analysis, Programming HW/SW Systems, Data Struct and Algorithms

Skills

Languages: SystemVerilog, Verilog, VHDL, C/C++, Java, RISC-V Assembly, Python, MATLAB

Digital Design: RTL Design, Finite State Machines, Computer Architecture, FPGA Design

Verification: ModelSim, Cadence Xcelium, RTL Simulations, Waveform Debugging

Tools: Intel Quartus Prime, Cadence Innovus, Git, VS Code, Linux

Hardware: Intel DE10-Lite FPGA, Arduino, Oscilloscopes, Logic Analyzer

Certifications: OSHA, CITI Human Subject Research

Organizations: SiliconJackets, GT Boxing Club (Secretary), Beta Chi Theta National Fraternity

Languages: English (fluent), Tamil (fluent), Spanish (beginner)

Projects

FPGA UART Communication & DSP Platform

May 2026 - June 2026

- Designed and implemented a modular SystemVerilog UART communication platform featuring UART TX/RX modules, FIFO buffering, command parsing, and FSM controller, supporting **115200 baud** and meeting **50MHz timing**.
- Developed a DSP subsystem which includes a moving average filter, modifiable clock divider, and 7-segment display driver, utilized **503 logic elements** and **282 registers** for the full design, achieving **125Mhz Fmax** and **+12.0ns setup slack**.
- Verified RTL functionality using ModelSim and UART with CP2102 USB-to-UART interface/Arduino IDE.

8-Bit CPU

June 2026 - July 2026

- Created a SystemVerilog 8-bit processor datapath implementing a program counter, instruction memory/decoder, register file, arithmetic logic unit (ALU), and FSM controller with 4 general-purpose registers meeting **50MHz timing**.
- Implemented **five logic instructions** (ADD, SUB, AND, OR, XOR) with register writeback and instruction execution.
- Verified RTL components using ModelSim and DE10-LITE hardware, utilized **70 registers**, **150 logic elements**, achieving **105Mhz Fmax**, **+10.5ns setup slack**, and **0 TNS**.

Digital ASIC Calculator Chip Design

January 2026 - March 2026

- Implemented SystemVerilog 64-bit SRAM-based arithmetic calculator with SRAM-based operand storage and FSM, combining upper and lower 32-bit operations with carry propagation,
- Verified the design using Xcelium, SimVision, and Innovus, with a driver, monitor, sequencer, scoreboard, 3 assertions, and **400 constrained random transactions**, achieving **99.95% coverage/100% assertion coverage**.

Experience

RevSync | Atlanta, GA

January 2026 - Present

Lead Hardware Engineer

- Contributed to initial design and hardware requirements while testing sensing, compute, and vehicle interfaces.
- Co-presented RevSync at GT Klaus Startup Challenge, communicating technical direction and roadmap, achieving an acceptance to CREATE-X LAUNCH.

Cultivate Lab | Athens, GA

January 2025 - January 2026

Undergraduate Researcher

- Co-authored 2 peer-reviewed engineering education publications (FIE 2025 AND ASEE 2026) examining university student mental health and coping strategies.
- Screened 5,000+ peer-reviewed articles using Rayyan to extract and synthesize data from 35 qualifying studies.

